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Project Title:	Molecular Characterization of Transmissible Chromosome Aberrations Produced by Ions of Intermediate and High Atomic Number: grant 80NSSC21K0679		
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Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:	**TechPort:**		No
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Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
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Key Personnel Changes/Previous PI:	December 2021 report: no changes.		
COI Name (Institution):	Yu, Yongjia Ph.D. (University of Texas Medical Branch)		
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	NOTE: Continuation as directed research of "Molecular Characterization of Transmissible Chromosome Aberrations Produced by Ions of Intermediate and High Atomic Number," grant NNX15AG74G; Principal Investigator Dr. Michael Cornforth continues as PI. Risks associated with crewed space flight involve extremely low fluences of mixed-field galactic cosmic radiation (GCR), and secondary neutrons formed through the interaction of lighter ions with the structure of spacecraft. It is simply not possible to duplicate this environment in any biologically meaningful experimental setting on Earth. The best solution to this problem is to simulate key aspects of the deep space environment by focusing on “representative” aspects of GCR. For obvious reasons, it is not feasible to measure directly biological risk to astronauts. Thus, we are forced to employ experimental model systems that mimic central aspects of the deep space radiation environment, and which employ surrogate biological endpoints of principle relevancy to biological endpoints of concern, like cancer.		

Task Description:	Most all ionizing radiation (IR)-induced chromosomal aberrations are exchanges that occur between different chromosome (e.g., translocations), or between regions within the same chromosome (e.g., inversions). Beyond the basic investigative platform that it provides to the science of radiation biology, knowledge concerning the genesis of IR-induced breakpoint junctions carries specific potential benefits. As for NASA and its priorities are concerned, a more detailed view of the enzymology of the rejoining reactions themselves would be of immense value to modelers, particularly as it relates to ultra-low dose risk assessment. Another benefit lies in the prospect that changes in radiation quality will influence the molecular structure of IR-induced breakpoint junctions. Colonies derived from the single-cell expansion of irradiated human telomerase reverse transcriptase (hTERT)-immortalized BJ-1 fibroblast cultures form the basis of this system. These cytogenetically characterized clones form the starting material for subsequent sequencing analysis. Specific Aims Aim 1: Determine if analysis of IR-induced chromosome aberration (CA) breakpoint junctions suggest different molecular pathways are involved, with respect to high- versus low Linear Energy Transfer (LET) exposure. Aim 2: Determine if the sizes of co-deleted sequences at breakpoint junctions differ for those produced by high versus low LET IR; these may form the basis of an LET-dependent biomarker. Aim 3: Determine if the breakpoints associated with complex chromosomal exchanges are different than those associated with simple reciprocal exchanges that involve only two chromosomes. Aim 4: Examine the breakpoint junctions of exchange aberrations induced by medium-to-high energy neutrons. Aim 5: Characterize molecular differences between the breakpoints produced by different ultra-low dose ions.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research. This study represents a continuation of Chromosome Aberrations I (NNX15AG74G) project; the objectives and research goals remain largely unchanged. We seek fundamental information at the molecular level about the nature of chromosome aberration breakpoints induced by ionizing radiations (IR) that differ vastly in ionization density, comparing terrestrial gamma rays to those of much higher linear energy transfer rates. To accomplish this will require further development and refinement of an experimental model system for which we have previously established proof-of-principle efficacy. This includes implementation of advanced methods of massively parallel sequencing such as single molecule/real-time sequencing.
Research Impact/Earth Benefits:	Radiation-induced reciprocal chromosome translocations and inversions are particularly important in that regard as they relate to crewed space activities. In addition to causing cancer, their appearance also accompanies ongoing genome instability processes associated with their progression. The fact that these particular chromosome aberrations are transmissible (non-lethal) also makes them ideal candidate biomarkers of accumulated radiation exposure. We argue that molecular analysis of breakpoint junctions formed as the result of translocations and inversions is vital to understanding the process of exchange aberration formation, since it is here where underlying repair/misrepair pathways leave their “molecular fingerprints.” Regarding relevance to NASA’s concerns, the study of chromosome aberrations stands to tell us much about mechanisms underlying the cancer process itself. The relationship between particle energy/track structure and radiogenic changes to the genome represents an important first step in understanding 1) basic dose-response relationships at low fluences and 2) fundamental carcinogenic processes that may ultimately form the basis for subsequent mitigation strategies.
Task Progress:	The overall structure of this proposal may be described as follows. For Objective 1, single human cells, initially exposed to various radiations, were grown into colonies that were later randomly isolated and cryopreserved for subsequent analyses. The first stage of the analyses involved 24-color chromosome painting (mFISH) to identify those clones whose cells harbored microscopically visible chromosome rearrangements such as translocations. [The technique of directional genomic hybridization (dGH) was also proposed to identify chromosome inversions.] These clones, most of which have now been collected, constituted the starting material for use in Objective 2, which involved the DNA sequencing of breakpoint junctions associated with radiation-induced translocations and inversions. This current report centers on progress made toward Objective 2. We proposed that DNA sequencing of these breakpoints would likely uncover fundamental differences in the types of damage produced by common gamma rays compared to that produced by radiations encountered in deep space environments (HZE particles), specifically high energy 56Fe accelerated ions produced at the NASA Space Radiation Lab (NSRL) Brookhaven facility. We had earlier reported success in sequencing the breakpoint junctions of a reciprocal translocation in a clone whose single progenitor cell had been exposed to 137Cs gamma rays, which we also published (Cornforth et al., 2018). The sequencing approach used there involved the use of paired-end sequencing, a so-called short-read approach that was hampered by the ubiquitous presence of repetitive DNA sequences in the human genome. As such, this approach was deemed insufficiently robust for our purposes. In last year’s report (see https://), we proposed the use of various newer sequencing strategies designed to circumvent this problem, among them mate-pair sequencing, and single-molecule/real-time sequencing (SMRT). While both offered improvement over our original pair-end sequencing approach, the latter proved especially useful for our purposes. Using SMRT, we now report having successfully sequenced the translocation breakpoints in two additional clones whose progenitor cells were exposed to gamma rays, and two derived following exposure to 1.0 GeV/nucleon accelerated 56Fe ions. For example, clone K1-100F2 harbored a gamma ray-induced reciprocal translocation between chromosomes 10 and 15. As with all reciprocal translocations, two breakpoint junctions are involved. In this case, and following Sanger sequencing, the chromosome 10-15 junction involved 16 bases of microhomology, whereas the 15-10 side of the translocation involved a large 30 basepair deletion, along with a 44-nucleotide duplication. At this point in our investigation we have not fully sequenced enough clones to definitively conclude whether the molecular signatures of chromosome aberrations produced by low LET (linear energy transfer) gamma rays are fundamentally different than those produced following exposure to HZE (high energy) particles. Thus, analysis of additional clones will constitute our major effort going forward. Our immediate objective for the next evaluation period is to publish our sequencing findings

	for a representative sample of six clones, three each from the 137Cs-gamma ray and 56Fe-exposed cohorts, in order to determine how the LET of a radiation might affect the breakpoint junctions of transmissible chromosome exchanges. We have several additional clones in the pipeline for that are undergoing various stages in the sequencing process. We expect additional publications to follow as more clones are completely sequenced. We anticipate that the problematic issues attached to the analysis of certain clones will be alleviated though the use directional genomic hybridization, which we have been striving to include as a routine part of our analysis because of its specific ability to detect chromosome inversions. We will hopefully have more to say about dGH in subsequent progress reports. As before, the greatest impediment to progress relates to breakpoints occurring in highly repetitive regions of DNA. While adequate progress is now being made using SMRT, we are guardedly optimistic that a newer ion-pore sequencing approach developed by Oxford Nanopore will help to accelerate this rate, because of its ability to provide much longer reads. My graduate Student, Stephen Kunkel, has been experimenting with this approach in combination with an integrated real-time algorithmic platform (Readfish). He recently submitted to NASA an augmentation to grant 80NSSC21K0679 entitled: Adaptive nanopore sequencing to increase the sensitivity of detecting transmissible chromosomal aberrations induced by ionizing radiation. The combined platform offers distinct advantages over SMRT such as lower cost but, more importantly, it allows for much longer read lengths than SMRT, which is especially useful in situations like ours, where a high proportion of breakpoint assignments are likely to involve repetitive DNA. Reference: Cornforth MN, Anur P, Wang N, Robinson E, Ray FA, Bedford JS, et al., Molecular Cytogenetics Guides Massively Parallel Sequencing of a Radiation-Induced Chromosome Translocation in Human Cells. Radiat Res 2018; 190, 88-97.
Bibliography Type:	Description: (Last Updated: 01/14/2026)
Articles in Peer-reviewed Journals	Loucas B, Shuryak I, Kunkel S, Cornforth M. "Dose-dependent transmissibility of chromosome aberrations at first mitosis following exposure to gamma rays. I. Modeling and implications related to risk assessment." Radiat Res. 2022 Jan 14. Epub ahead of print. https://doi.org/10.1667/RADE-21-00180.1 ; PMID: 35030259 , Jan-2022
Articles in Peer-reviewed Journals	Shuryak I, Loucas BD, Cornforth MN. "Robbing Peter to Pay Paul: Competition for radiogenic breaks during rejoining diminishes curvature in the dose response for simple chromosome exchanges." Radiat Res. 2021 Aug 1;196(2):147-55. https://doi.org/10.1667/RADE-20-00253.1 ;PMID: 34019659; PMCID: PMC8440481 , Aug-2021